

Arghadeep Roy

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Education

- 2022– **Ph.D. Candidate**, *University of Oklahoma*, Norman,
Advisor: Prof. Michael Jablonski.
- 2019–2022 **M.Math**, *Indian Statistical Institute*, Bangalore
- 2016–2019 **B.Math**, *Indian Statistical Institute*, Bangalore

Research Interests

Differential Geometry, Geometric Analysis

Talks

- October 2025 *Betti Numbers under Ricci Curvature Bounds (Graduate student geometry and topology seminar)*
- April 2025 *Homogeneous Einstein and Ricci soliton metrics (Geometry and Topology Seminar)*
- March 2025 *Ricci soliton metrics and Homogeneous Spaces (Graduate student geometry and topology seminar)*
- April 2024 *Incompressible Surfaces in 3-Manifolds (Graduate student geometry and topology seminar)*
- May 2021 *The Kodaira Embedding Theorem (Master's Project Presentation at ISI)*
- March 2021 *Spinor Representation of the Lorentz Group*

Conferences and Workshops Attended

- August 2026 *Frontiers in Global Differential Geometry (James Madison University)*
- June 2026 *Summer Graduate School on Geometric Measure Theory (SLMath)*
- May 2026 *Pacific Northwest Geometry Seminar (University of Washington, Seattle)*
- April 2026 *Soul and Shape: Metric Structures and Geometric Analysis (University of Oklahoma)*
- April 2026 *Texas Geometry and Topology Conference (Texas Christian University)*
- November 2025 *Texas Geometry and Topology Conference (University of Texas, Dallas)*
- October 2025 *AMS Sectional Meeting: Special Session on Riemannian Geometry and Symmetry (St. Louis University)*
- September 2025 *Symmetry and Geometry at Oklahoma (University of Oklahoma)*

June 2025 *Midwest Panorama of Geometry and Topology (The University of Iowa)*
 February 2025 *Southern California Geometric Analysis Seminar (University of California, Irvine)*
 November 2024 *Texas Geometry and Topology Conference (Texas A&M University)*
 October 2024 *Redbud Topology Conference (OU)*
 April 2024 *Kansas Geometric Analysis Conference (Wichita State University)*
 February 2024 *Symmetry and Geometry in South Florida (Florida International University)*
 March 2023 *Redbud Topology Conference (OU)*
 March 2023 *Midwest Geometry Conference (Kansas State University)*
 January 2022 *IIT Bombay Geometric Analysis Seminar*
 August 2021 *6th Geometry-Topology Summer School (Feza Gürsey Institute (online))*
 July 2020 *5th Geometry-Topology Summer School (Istanbul Center for Mathematical Sciences (online))*
 December 2019 *Advanced Instructional School in Geometric Analysis (IIT Bombay)*

Teaching and Teaching Assistantships

Instructor

Summer 2026 *Precalculus and Trigonometry*
 Spring 2025 *Calculus I for Business, Life, and Social Sciences*
 Fall 2024 *Precalculus and Trigonometry*
 Spring 2024 *College Algebra*
 Spring 2023 *Functions and Modeling for Business, Life and Social Sciences*

TA

Fall 2026 *Differential and Integral Calculus II*
 Spring 2026 *Calculus and Analytical Geometry I*
 Fall 2025 *Calculus and Analytical Geometry I*
 Spring 2025 *Applied Modern Algebra*
 Summer 2024 *Linear Algebra I*
 Spring 2024 *Calculus and Analytical Geometry IV*
 Fall 2023 *Calculus and Analytical Geometry II*
 Fall 2023 *Linear Algebra I*
 Fall 2022 *Calculus and Analytical Geometry III*
 Fall 2022 *Introduction to Ordinary Differential Equations*

Service and Mentorship

2026 *Chair of the Graduate Advocacy Committee*

- 2026 *Directed Reading Program*
- 2025 *Vice-President of the Mathematics Graduate Student Association*
- 2025 *Organsied the Graduate student geometry and topology seminar*

Awards

- 2026 *Thomas J. Hill Graduate Fellowship (OU)*
- 2025 *C. Eugene Springer Mathematics Scholarship (OU)*
- 2025 *Summer Research Fellowship (OU)*
- 2019–2022 *M.Math Honors Fellowship*
- 2016–2019 *B.Math Honors Fellowship*

Languages

English, Bengali, Hindi

Technical Skills

C, C++, Octave, Python, R, Latex